Assignment 4: Text and Sequence Data

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1. Implemented in code.
2. Implemented in code.
3. Implemented in code.
4. Implemented in code.
5. The models show that with 100 samples, the pretrained embedding slightly outperforms the learned embedding because it does not need to learn word representations from limited data. As the training size increases, the learned embedding quickly becomes more effective. With 500 samples, it starts outperforming the frozen embedding as highlighted below. With 5000 samples, the learned embedding achieves a much higher accuracy. This demonstrates that pretrained embeddings are useful for very small datasets, but learned embeddings become much better once the model has enough data to learn specific word representations.

Training Size	Learned Embedding Accuracy	Pretrained Embedding Accuracy	Best Model
100	0.5028	0.5070	Pretrained
500	0.5085	0.5003	Learned
1000	0.5132	0.5021	Learned
5000	0.7641	0.5186	Learned